BrandGuard Preflight

Product plan | Inkloom AI problem statement | 25 October 2026 | Author: Somnath Jamadar

Team: Somnath (Team Lead, MERN Full Stack Developer) | Falguni Sharma (MERN Full Stack Developer) | Om Apar (AI/ML Specialist)

Preflight checks content before it goes out. Every claim in a draft is compared with the company's own approved guidelines, facts and past content. When we can prove a problem, we show the proof and suggest a fix that keeps what the writer meant. When we can't prove it either way, a person decides.

1. Problem definition and evidence

Most growing companies have brand guidelines, but in practice nobody checks content against them. Writing now comes from employees, agencies and AI tools, and mistakes like a wrong claim, an old price, off-tone wording or an off-brand image often reach the public before anyone spots them. The surveys below come from vendors and trade press, so read them as a rough guide, but they all tell the same story:

- 85% of organisations have brand guidelines but only about 30% enforce them consistently; 77% regularly deal with off-brand content (Marq).
- 47.1% of marketers meet AI inaccuracies several times a week and 36.5% say incorrect AI content has already gone public (NP Digital, 565 US marketers).
- 50% of consumers prefer brands that avoid GenAI in consumer-facing content; Gartner advises labelling AI use (1,539 US consumers).

Most tools look at how a piece of writing *sounds*. We also check what it *says*: is each claim true, is it current, and does it match what the brand has already published?

1.1 How we read the problem statement

The brief is short, but almost every phrase in it is a design requirement. Below, we turned each phrase into what it really asks for and how Preflight responds.

Phrase in the brief	What it really demands	Our design response
Content from multiple employees, agencies and AI tools	Many authors, no shared memory, rising volume; AI makes errors faster	One gate that every source must pass, including an API for other AI tools
Incorrect claims, outdated information, inconsistent language, off-brand visuals	Facts have a shelf life; errors must be caught before publication, not after	Claim Shelf-Life, contradiction check against past content, visual and accessibility checks
Check against approved guidelines, facts and assets	The ground truth must be the organisation's own approved material, not the model's opinion	Evidence-bound findings: no source means no violation
Explain issues and suggest corrections that preserve purpose	Feedback must teach, and the creator's purpose has to be captured first	Evidence card; Intent Lock with a meaning check on every fix
Creators, managers and authorised reviewers collaborate and track changes	Roles, comments, approvals and an accountable record	Role-based workflow, version diffs, Brand Passport
Insights into repeated mistakes and unclear guidelines	The guidelines themselves are a product that needs debugging	Insights dashboard and Guideline Debugger

Balance automation with human control; protect confidential material; reliable, explainable results	Explicit rules for what software may decide, and a security-first design	Decision matrix, redaction preview, rules-only mode, disclosure of AI use

1.2 Root causes behind the failures

1. **Guidelines are written for people, not machines.** A rule like "keep a friendly but professional tone" can't be checked the same way twice, so enforcement depends on who happens to review.
2. **Facts have no owner and no expiry date.** Prices, customer counts and policies change, but old copies stay in decks, posts and web pages and nobody is told.
3. **Feedback comes late and without a reason.** Problems show up after publishing or after several review rounds, so people start working around review.

1.3 Design tensions we must resolve

Tension	Our position
Speed vs control	Automate only what can be proven exactly; route everything unproven to a person with the evidence attached
Catching more vs being trusted	Prefer fewer, sourced findings over many guesses; a tool that cries wolf gets ignored
AI help vs confidentiality	Local-first design, redaction preview before any external call, and a mode with the model switched off
Strict rules vs creative freedom	Suggestions are opt-in and overrides need a reason; repeated overrides signal an unclear rule to fix, not a person to blame

1.4 A typical failure, step by step

Step	What happens today	Where it goes wrong
1	The returns policy changes from 14 days to 30 days. The policy page is updated.	Nobody is told that older posts, decks and FAQs now contradict it.
2	A new agency writer drafts a promotional post from an old brief that says "14 days".	The guidelines don't mention return periods, and the writer has no way to look up the current fact.
3	An AI tool rewrites the post and adds "the best returns in India".	The new claim has no evidence behind it, and the tool sounds confident.
4	The brand manager skims the post, likes the tone and approves it.	Checking facts and claims was never part of the review, and she is reviewing 40 other items.
5	The post goes live. A customer quotes it at the support desk.	The error is found by a customer, after the damage is done.

Preflight steps in at step 2 and 3: it shows the writer the current policy next to the draft, and it asks legal about the superlative before the post goes anywhere near the brand manager.

1.5 What a good solution has to do

1. Catch problems before publication, not after.
2. Base every judgement on the organisation's own approved material, so results can be checked.
3. Explain each issue and offer a fix that keeps the writer's purpose, so people learn and don't work around it.
4. Hand anything uncertain or risky to the right person, with the evidence ready.
5. Keep confidential material inside the organisation and say clearly when AI or an outside service is used.

6. Show where the guidelines themselves are the problem, so the same mistake stops repeating.

2. Target users, needs and organisational setting

We are designing for a growing company of roughly 100 to 2,000 people. Marketing, sales, support and regional teams all publish, agencies and AI tools help out, and a small brand team of 2 to 8 people owns the guidelines but can't read everything.

Persona	Core need	Main fear
Creator (marketer, sales rep, agency writer)	Fast feedback with proof; fixes that keep my message	Vague rejections
Brand manager	Catch issues before publishing; stop repeat mistakes	Becoming the manual bottleneck
Authorised reviewer (legal, product, compliance)	Only items needing expert eyes, with evidence attached	Approving something unsupported
Knowledge owner / admin	One place to keep facts current	Stale or leaked material
Leadership	Proof that risk is falling	A public brand incident

3. Research, interviews, observations and surveys

So far our evidence is secondary: published surveys and standards, listed in section 1 and in the table below. We have planned interviews, a survey and some observation for the next stage, but we won't report any results until we have actually collected them.

3.1 Secondary evidence collected

Source	Finding	Used for
Marq (formerly Lucidpress) brand consistency research	85% of organisations have guidelines, about 30% enforce them consistently, 77% face off-brand content	Problem, rule enforcement
NP Digital survey, 565 US marketers (Feb 2026)	47.1% meet AI inaccuracies weekly; 36.5% say incorrect AI content went public; 70%+ spend 1 to 5 hours a week fact-checking	Evidence-bound checking, human review
Gartner survey, 1,539 US consumers (Mar 2026)	50% prefer brands that avoid GenAI in consumer-facing content; labelling AI use advised	AI-use disclosure
Berkeley SCET overview of hallucinations	Factuality and faithfulness hallucinations are distinct failure types	Grounding every finding in a source
WCAG 2.2 (W3C), OWASP Top 10 for LLM Applications	Accessibility and LLM security standards	Accessibility checks, prompt-injection defence

3.2 Primary research plan

Method	Who and how many	What we want to learn	Findings
Interviews	5 to 8: brand or marketing managers, creators, one legal or compliance reviewer, one agency writer	How content is checked today, last off-brand incident, what would make them trust automated feedback	to fill

Survey	30+ marketers and content creators	Frequency of off-brand or wrong content, time lost in review, share of AI-assisted content	to fill
Observation	2 to 3 real review cycles	Rounds taken, who comments on what, which issues repeat	to fill
Guideline audit	3 to 5 public brand guidelines	How ambiguous the rules are; which can become automated checks	to fill
Prototype test	5 users on the prototype	Are proofs clear, are fixes acceptable, false-positive rate	to fill

What we want to test: (1) creators accept feedback more readily when they can see the source passage and the fix keeps their message; (2) contradicting past published content is common and rarely checked; (3) reviewers spend most of their time on a small share of issues, mainly claims and facts; (4) many repeated mistakes come from unclear guidelines, not careless writers; (5) teams will only adopt AI checks if people keep final approval and the tool shows its evidence.

3.3 Research by team member and domain

Each of us researches the area we own. The last column is for each person's own notes and links, so only work that was really done appears there.

Somnath: backend, security and workflow

Research focus	What the evidence says	Design decision	Member notes
Role-based review workflows and audit logging in content approval	Review is where brand risk is caught; guidelines are rarely enforced (about 30% consistently)	Four roles, versioned drafts, immutable audit log	to fill
Securing confidential uploads: tenant isolation, encryption, retention	Marketers cite privacy and stolen data among top GenAI concerns (Statista, June 2025)	Per-organisation isolation, encrypted storage, deletion controls	to fill
LLM application risks such as prompt injection and data leakage (OWASP LLM Top 10)	Uploaded documents can contain hostile instructions	Treat uploads as data; redaction gateway before external calls	to fill
Content hashing for tamper-evident records	Standard hashing (SHA-256) is enough to detect later edits	Brand Passport stores a content hash per approved version	to fill
API design and integration points: OpenAPI contracts, Slack or Teams notifications, Docs and CMS hooks	Teams need feedback where they already write	Versioned REST API; Agent Gate API for external AI tools	to fill

Falguni Sharma: frontend and UX

Research focus	What the evidence says	Design decision	Member notes
How reviewers consume feedback: inline highlights, side-by-side comparison, threaded comments	Over 70% of marketers spend 1 to 5 hours a week fact-checking AI content (NP Digital)	Evidence card with draft, proof and verdict side by side	to fill
Accessibility standards: WCAG 2.2 contrast, alt text, heading order, plain language	WCAG AA requires 4.5:1 contrast for normal text	Accessibility checks and an accessible UI of our own	to fill

Existing brand-governance and style tools: what their review screens show and miss	Most check style and assets; few show source-linked proof	Proof panel and comparison views as the core screen	to fill

Om Apar: machine learning

Research focus	What the evidence says	Design decision	Member notes
Claim extraction and claim-to-source matching using embeddings and natural-language inference	LLMs produce plausible but wrong content; 36.5% of marketers saw incorrect AI content go public (NP Digital)	Model compares only against retrieved passages; every verdict needs a source	to fill
Hallucination and faithfulness: grounding outputs in provided context	Factuality and faithfulness hallucination are distinct failure types (Berkeley SCET overview)	Ungrounded output is shown as a question, not a violation	to fill
Tone and voice modelling from approved examples; contradiction detection against past content	Tone judgement is the least reliable part of automated review	Tone is advisory; contradictions and facts are the blocking checks	to fill
Image checks: OCR, colour palette extraction, logo matching, contrast ratio	Visual drift is common and often caught only in design review	Basic image checks first; logo matching as stretch	to fill
Evaluation: planted-issue test set, precision, recall, confidence calibration	Accuracy, not speed, is the competitive advantage (NP Digital)	60-draft seeded test set and published metrics	to fill

Sources used in this plan: Marq via Emulent; NP Digital survey via Martech Edge; Gartner, March 2026; Berkeley SCET on hallucinations; Statista, B2B marketer GenAI concerns; WCAG 2.2 (W3C) and OWASP Top 10 for LLM Applications are cited as standards.

4. Proposed solution, value proposition and workflow

Every draft has to pass three tests, and for each result Preflight shows the proof behind it.

Test	Question	Checked against
Truth	Is every claim consistent with approved guidelines, facts and the brand's own published history?	Uploaded guidelines, fact sheets, claim evidence, past approved content
Risk	How could this be misread or quoted against us?	Hostile-reader pre-mortem; confidential-term watchlist
Intent	Do suggested fixes still say what the creator meant?	Creator's stated purpose, audience and must-say points

Evidence card: nothing is flagged without all three parts

Part	Shows
Draft	The exact sentence, highlighted
Proof	Matching passage from your uploaded source or past content, with file name, page and version
Verdict	Supported, Contradicted, Outdated, Not found or Unclear, plus a confidence level

Value proposition by stakeholder

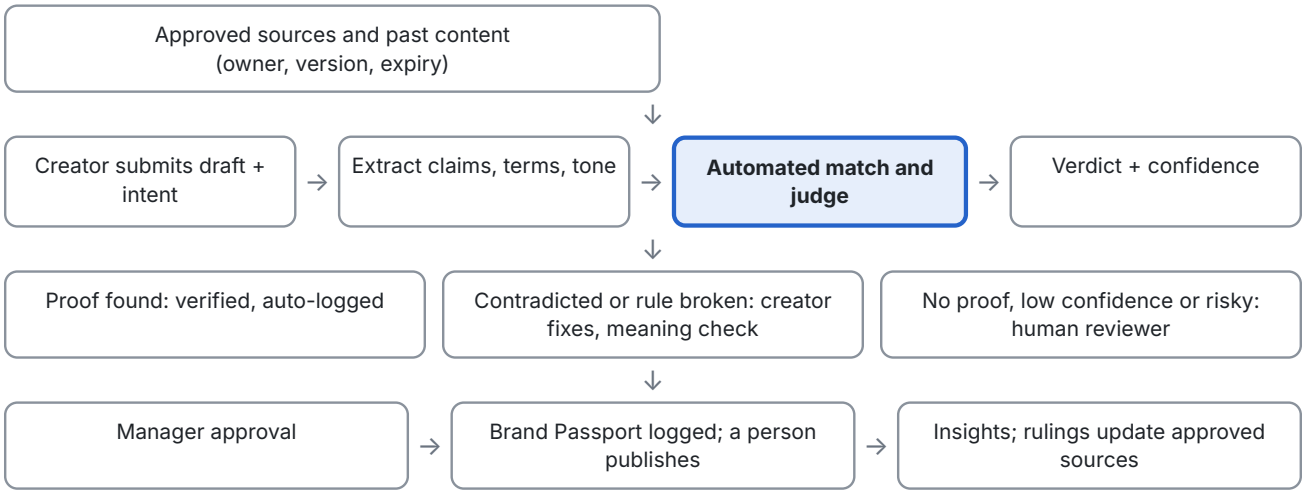
Stakeholder	Value
Creators	Instant feedback with proof; fewer rewrite rounds; fixes that keep their message
Brand managers	Fewer late catches; a view of where guidelines are unclear
Reviewers	Only unproven or high-risk items, with evidence already attached
Organisation	Lower risk of public errors, contradictions and leaks; an audit record for every published asset

Who decides: automation, creator or human

Situation	System does	Who acts
Claim matches an approved fact	Marks verified, logs it	Nobody
Exact rule broken (banned term, low contrast, no alt text)	Flags with rule and suggested fix	Creator accepts, edits or dismisses
Draft contradicts a source (high confidence)	Shows both passages and a correction	Creator, then manager
Source expired, or two sources disagree	Does not guess; asks the owner	Knowledge owner
No proof found	Shown as a question, never a violation	Right reviewer (legal, product)
Low confidence or risky topic (legal, health, finance, comparisons)	Packs evidence and queues it	Authorised reviewer
Reviewer rules on an item	Records reason; offers to save as approved fact or rule	Owner confirms

End-to-end workflow

Only the highlighted step runs on its own. Every other step is a person deciding something, or the system writing it down.



What a finding looks like

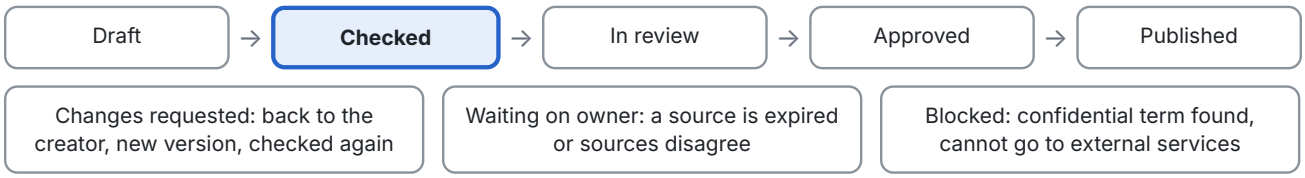
This is the card a creator sees for the returns example. Everything on it comes from the uploaded sources or from the creator's own intent.

Finding 1 of 4: Factual claim — **Contradicted** Confidence 92%

Draft	"Free returns within 14 days."
Proof	Returns Policy, version 2, page 3: "Customers may return items within 30 days of delivery." (owner: Support, expires 31 Mar 2027)
Why flagged	The draft states a return period that differs from the approved policy.
Suggested fix	"Free returns within 30 days."
Meaning check	Kept: free returns, a time limit. Changed: 14 days to 30 days. Dropped: nothing from the creator's must-say list.

Accept fix | Edit | Dismiss with reason | Ask the owner

Review lifecycle



A creator's journey

1. Priya, a marketer, uploads a draft and fills in a short intent form: purpose, audience and two must-say points.
2. Preflight returns four findings in under a minute. Two are verified, one is a contradiction with the returns policy, one is a superlative with no evidence.
3. She accepts the fix for the policy error and edits the suggested rewrite of the superlative.
4. The "best returns in India" line has no evidence, so it is routed to the product and legal reviewers as a question. She keeps working on the rest.
5. The legal reviewer rules that the claim can't be used. The ruling is saved so the next writer who tries the same claim is told straight away.
6. Her manager approves the new version, a Brand Passport is saved, and Priya publishes.

What is different from typical approaches

This compares general kinds of tools, not specific vendors. Before the build we will check it against a few real products and fix anything we got wrong.

Typical approach	Usually good at	Usual gap	What Preflight adds
Style and grammar checkers	Spelling, tone, readability	Do not know whether a claim is true or current	Claim checking against approved facts with expiry dates
Brand portals and asset libraries	Storing logos, templates and rules	Rules are not checked automatically on each draft	Per-draft review with source-linked findings

General AI chat used as a reviewer	Fluent suggestions	May invent issues; no source, no audit trail, data leaves the company	Evidence-bound output, redaction, audit log, human routing

What makes us different, in one line: instead of judging how content sounds, Preflight checks what it says against the company's own approved facts and its own published history, and every human ruling becomes knowledge it can reuse. We haven't seen these ideas emphasised together:

- Findings that can't exist without a cited source passage: no source, no violation.
- Contradiction checks against the brand's own past content.
- Claim Shelf-Life: facts have owners and expiry dates, and the owner is warned before one lapses.
- Intent Lock: every fix comes with a note on what it keeps and what it drops.
- Guideline Debugger and Rule Simulator: guidelines get tested and improved like any other asset.
- Brand Passport: a tamper-evident record for each approved asset, including whether AI was used.

5. Planned features, technology stack, integrations and data sources

5.1 Core features

Capability	What it does	Phase
Knowledge base	Import or define guidelines (PDF, DOCX, Markdown), approved facts, terminology, claim evidence, reusable assets and past approved content; owner, version and expiry on every item	MVP
Content review	Text, DOCX and PDF; PPTX and images next; findings mapped to page, slide or region	MVP
Detection	Tone, terminology, messaging, facts, claims, visual usage, accessibility	MVP
Evidence cards and suggestions	Why flagged, which source, suggested fix; accept, edit or dismiss	MVP
Review workflow	Roles, comments, approvals, assignments, revision history with diffs	MVP
Insights dashboard	Repeated mistakes, unclear rules, time to approval, by team and rule	Phase 2
Privacy and disclosure	Role-based access, audit log, redaction preview, visible AI-use notice	MVP

5.2 Unique features we are implementing

#	Feature	What it does and why it is different	Phase
1	Evidence-bound findings	A finding must cite a passage in the organisation's own uploads. No source means "not found" and a human is asked, so the tool never invents a violation.	1
2	Self-contradiction check	Compares a draft with everything the brand already published, not just the rulebook (for example 14-day returns vs the 2025 policy of 30 days).	1
3	Claim Shelf-Life	Every approved fact has an owner and expiry date. Owners are warned before it lapses; drafts using an expired fact are flagged "outdated".	1
4	Intent Lock with meaning check	Creator states purpose and must-say points; every fix shows what was kept and what was dropped.	1

5	Hostile-reader pre-mortem	Shows the single line most likely to be quoted against the brand and why, from regulator, journalist, competitor and screen-reader viewpoints.	1
6	Rulings become knowledge	A reviewer's decision can be saved as an approved fact or rule, so the same question is resolved automatically next time.	1
7	Guideline Debugger	Highlights rules that are often overridden or disputed and suggests clearer wording, pointing to where the guidelines themselves are unclear.	1
8	Brand Passport	Each approved asset gets a record: content hash, checks run, rules version, approvers and whether AI was used. Tampering after approval is detectable.	1
9	Redaction preview and zero-egress mode	Confidential terms are replaced with placeholders before any external AI call and the user sees what would leave. Rules-only mode runs with the model off.	1
10	Confidential-term tripwire	Blocks drafts that mention embargoed names (unreleased products, client names) before the announcement.	1
11	Rule Simulator	Before changing a guideline, replay it on the last 100 drafts to see how many would be flagged.	2
12	Agent Gate API	Other AI tools and agencies must obtain a Preflight result before publishing, so AI content cannot skip review.	2
13	Voice Fingerprint	Learns the brand's tone from approved posts and shows a tone-drift meter.	2
14	Multilingual consistency	Checks that a Hindi or Marathi version keeps the same claims as the English original.	2
15	Accessibility auto-suggest	Drafts alt text, plain-language rewrites and contrast fixes for the creator to approve.	2
16	Trust Report export	One PDF per asset with evidence, approvals and AI-use disclosure, for audits.	2

5.3 Technology stack and team ownership

We are using the MERN stack (MongoDB, Express, React, Node) plus a Python service for the ML work. Every part has one owner.

Somnath: Backend and APIs (Node.js, Express, MongoDB)
REST API design and OpenAPI contract, versioned endpoints, integrations and Agent Gate API, auth and roles, workflow engine, knowledge-base storage, versioning and audit log, file handling, security and tenant isolation, redaction gateway, ML service orchestration, insights aggregation

Falguni Sharma: Frontend (React)
Upload and review screens, evidence cards, diff and comparison views, comments and approval UI, dashboard, accessible and responsive interface, calls Somnath's API

Om Apar: Machine learning (Python)
Claim extraction, embeddings and retrieval, claim-to-source matching, tone and contradiction models, intent meaning check, pre-mortem, OCR and image checks, evaluation and test set

Layer	Choice	Owner	Why
Front end	React with side-by-side evidence and diff views	Falguni	Highlights, proof panel, comments, revisions
API layer	REST with an OpenAPI spec; versioned endpoints	Somnath	Clear contract between UI, backend and external tools

Server	Node.js and Express	Somnath	Workflow, roles and orchestration in one language with the UI
Database	MongoDB: users, roles, comments, versions, audit log, passports	Somnath	Flexible documents for findings and revisions
Auth and security	JWT or OIDC, role-based access, per-organisation isolation, encryption, upload scanning	Somnath	Protects confidential brand material
File storage	Encrypted object storage (S3-compatible)	Somnath	Uploaded sources, drafts, exports
Parsing and OCR	PyMuPDF, python-docx, Tesseract	Om	Text, page and slide positions from documents and images
Retrieval	Sentence embeddings with a vector index (FAISS or Chroma), stored with file, page, version, expiry	Om	Every finding links to a source passage
Matching and judging	Rules first, then NLI-style semantic matching; confidence scoring	Om	Fewer false alarms, explainable verdicts
Language model	Configurable provider with no-training, no-retention terms, or a small local model; can be turned off	Om	Rewrites, intent check, pre-mortem
Image checks	OpenCV: palette, logo matching, text-over-image contrast	Om	Visual consistency and accessibility
ML service	FastAPI wrapper, called only by the Node backend	Om (service), Somnath (calls)	Clean boundary between app and models

One rule we are keeping: the browser never talks to the ML service directly. Falguni's UI calls Somnath's API, and the backend applies access rules and redaction before it calls Om's service.

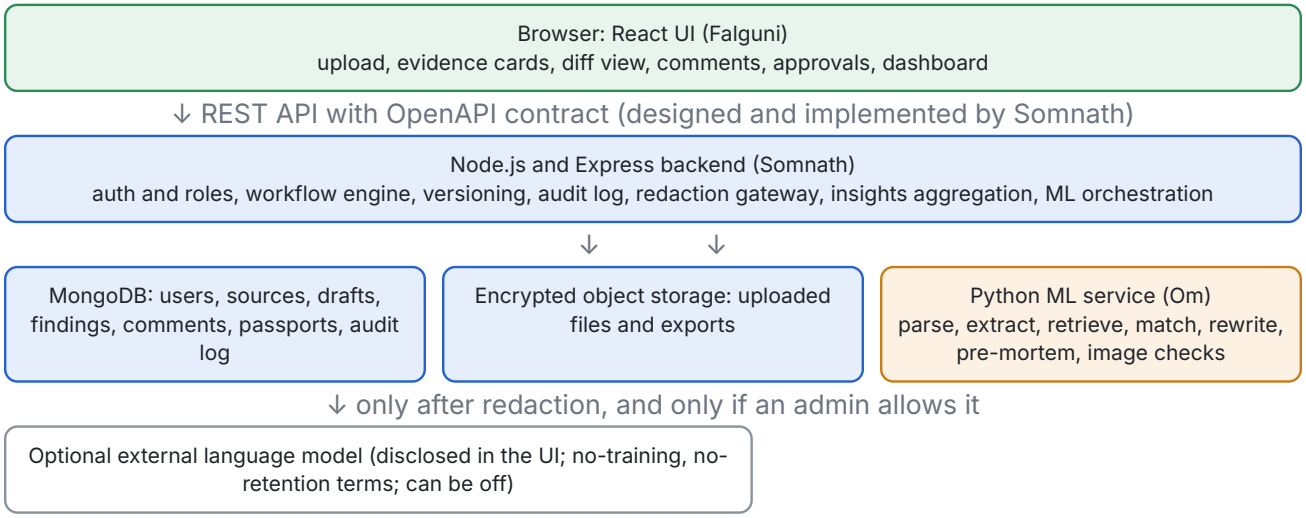
5.4 Integrations

Integration	Purpose	Phase
Web app with upload and paste	Main entry point for creators and reviewers	MVP
Slack or Microsoft Teams	Review and approval notifications where teams already work	Phase 2
Google Docs or CMS plug-in	Check drafts where they are written	Phase 2
Agent Gate API	Lets other AI tools and agencies obtain a Preflight result before publishing	Phase 2
Model provider (optional, external)	Rewrites and meaning checks, with no-training and no-retention terms; disclosed to users and can be switched off	MVP

5.5 Data sources

We only use material the organisation gives us: brand guidelines, approved fact sheets, glossary and terminology, claim-evidence records, the asset library (logos, palette, templates) and past approved content. Nothing is pulled from the web unless an admin turns that on. For the demo we will use a made-up company with synthetic documents, so no real confidential material is involved. Over time, reviewer rulings and decision logs from inside the product become another source for insights.

5.6 System architecture



Colours show who owns what: green is frontend, blue is backend and APIs, orange is machine learning. The browser never calls the ML service directly.

5.7 Data model (MongoDB collections)

Collection	Key fields	Purpose
organisations, users	orgId, role (creator, manager, reviewer, owner), permissions	Tenant isolation and role-based access
sources	file, type, owner, version, effective date, expiry date, status	Approved guidelines, facts, terminology, assets
passages	sourceId, version, page, text, embeddingRef	Retrieval units that findings cite
facts	statement, value, evidence link, owner, expiry, supersedes	Claim Shelf-Life and approved claims
drafts, draftVersions	content, format, intent (purpose, audience, must-say), author, contentHash	Revision history and diffs
findings	draftVersion, span, type, verdict, confidence, proofPassageId, suggestedFix, status	Evidence cards
comments, decisions	thread, author, action (approve, request changes, override), reason	Collaboration and accountability
passports, auditLog	contentHash, checks run, rulesVersion, approvers, aiUsed; append-only events	Brand Passport and traceability

5.8 Main API endpoints

Endpoint	Purpose
POST /sources, GET /sources/:id	Upload and define approved material; list versions and expiry
POST /drafts, POST /drafts/:id/review	Submit a draft with intent and run the review
GET /drafts/:id/findings, PATCH /findings/:id	Read evidence cards; accept, edit or dismiss with a reason
POST /drafts/:id/comments, POST /drafts/:id/approve	Collaboration and approval steps by role
GET /passports/:id	Verify the record of an approved asset
GET /insights/rules, GET /insights/mistakes	Governance insights and unclear-rule candidates

POST /gate/check	Agent Gate: external AI tools request a Preflight result before publishing

Internal ML endpoints, called only by the backend: /extract-claims, /match, /rewrite, /premortem, /image-check, /detect-entities.

5.9 ML pipeline, models and evaluation

Stage	Approach	Why
Parse	PyMuPDF and python-docx for text and page positions; Tesseract OCR for images	Findings point to the exact page or region
Extract claims	Rules and a small language-model step to pull out numbers, names, product terms and promotional claims	Focus on checkable statements first
Retrieve	Multilingual sentence embeddings with a vector index (FAISS or Chroma); passages stored with version and expiry	Candidate proofs for each claim; supports Hindi and Marathi later
Judge	Exact rules first; then an entailment-style comparison between claim and retrieved passage; verdict and confidence	Supported, Contradicted, Outdated, Not found or Unclear
Suggest	Language model drafts a fix constrained to the retrieved facts and the creator's intent; meaning check compares original and fix	Fixes that preserve purpose
Pre-mortem	Structured prompts from regulator, journalist, competitor and screen-reader viewpoints; outputs the riskiest line	Risk test beyond rules
Image checks	OpenCV for palette and contrast; logo template matching as stretch	Visual consistency and accessibility

Evaluation. We will use a seeded test set of about 60 drafts with planted issues and report recall, precision, how often findings cite a source, how many contradictions we catch, and whether our confidence is calibrated (does "90% sure" really mean about 90% correct?). Findings that reviewers mark as wrong go back into the test set. Low-confidence results never block anything; they go to a person.

5.10 How each unique feature works technically

Feature	Technique	Owner
Evidence-bound findings	Every finding stores a proofPassageId. The API refuses to save a finding of type violation without one; otherwise it is stored as a question.	Somnath, Om
Contradiction against past content	Past approved content is embedded and indexed like any other source. Nearest passages are compared with the claim using an entailment-style model.	Om
Claim Shelf-Life	Expiry date on every fact. A scheduled job warns owners 30 days ahead; at review time an expired fact returns the verdict Outdated.	Somnath
Intent Lock	Structured intent form. Each fix is checked for coverage of the must-say points and for similarity to the original meaning, and the result is shown as kept, changed and dropped.	Om, Falguni
Pre-mortem	Several viewpoint prompts (regulator, journalist, competitor, screen-reader user) with a fixed JSON output. We keep the single highest-risk line and its reason.	Om
Brand Passport	SHA-256 of the approved content plus checks run, rules version, approvers and AI-use flag, stored in an append-only collection.	Somnath

Redaction and tripwire	Org watchlist and entity detection replace confidential terms with placeholders before any external call; embargoed terms block the draft until the embargo date.	Somnath, Om
Guideline Debugger	Per-rule counts of flags, overrides and disputes. Rules with a high override rate are ranked; a suggested rewording is proposed for a person to approve.	Somnath, Falguni

5.11 Confidence thresholds and routing

Situation	What the system shows	Who handles it
Exact rule matched	Finding with the rule and a fix	Creator
Claim matched to a passage, confidence 0.85 or higher	Supported or Contradicted, with proof	Creator, then manager if Contradicted
Confidence between 0.60 and 0.85	Unclear, shown as a question	Creator; reviewer if the topic is risky
Confidence below 0.60, or no passage found	Not found, shown as a question	Reviewer or knowledge owner
High-risk topic (legal, health, finance, comparisons)	Always queued, whatever the confidence	Authorised reviewer

These numbers are starting values. We will tune them on the test set and publish the final values with the results.

5.12 Threat model and controls

Threat	Control
Hostile instructions hidden in an uploaded document (prompt injection)	Uploaded text is treated as data. The model gets no tools, outputs must match a fixed schema, and anything that doesn't is discarded.
One organisation seeing another's data	orgId on every record and query, separate vector index per organisation, checks enforced on the server.
Confidential text sent to an outside model	Redaction gateway with a preview, a per-organisation allow-list for external services, and a rules-only mode.
Someone approving beyond their role	Role checks on the server for every approval; overrides need a written reason and are logged.
Changes to approved content or to the log	Content hash in the Brand Passport; append-only audit log that even admins cannot edit.
Malicious or oversized uploads	File-type allow-list, size limits, virus scan, parsing in an isolated process.
Made-up findings from the model	No source, no violation; schema validation; confidence thresholds; low confidence goes to a person.
People trusting a green tick too much	Every report states "checked against these sources only" and lists what was not checked.

5.13 Testing, performance and deployment

- **Testing.** Unit tests for every rule; a golden test set for the ML pipeline; API tests against the OpenAPI contract; one end-to-end test that replays the demo story.
- **Performance goal.** A one-page draft reviewed in under 30 seconds on a laptop-class machine, with the model off or on a small local model.

- **Deployment.** Docker Compose for the demo (web, API, ML service, MongoDB). Settings come from environment variables and no secrets are stored in the repository.
- **Accessibility of our own UI.** Keyboard navigation, sufficient contrast and automated checks with axe.

5.14 Choices we considered

Decision	Chosen	Why, and the alternative
Database	MongoDB	The team knows MERN, and findings and revisions are document-shaped. PostgreSQL with pgvector would also work.
Vector search	FAISS or Chroma inside the ML service	Runs locally, so embeddings never leave the organisation. A managed vector database would add cost and data exposure.
Judging method	Rules first, then an entailment-style model, then a language model for rewrites	Predictable and explainable. A single large model prompt is faster to build but harder to trust.
Model hosting	Optional external provider with no-training terms, or a small local model	Gives a way to run with no external calls at all.

6. Responsible design: privacy, security and human oversight

People make every important decision, and whenever the system says something, it has to show its evidence.

Area	Commitment	How
Human oversight	AI never publishes, approves or rewrites on its own	Fixes are opt-in; high-risk topics need a named reviewer; overrides need a written reason
Explainability	No finding without a source	Every finding shows passage, file, page, version, severity and confidence; ungrounded output is labelled a question
Reliability	Fewer false alarms beats more alarms	Rules first, model second; thresholds; wrong findings feed a test set, not a hidden retrain
Privacy and security	Brand material is confidential by default	Tenant isolation, encryption, least-privilege access, audit log, configurable retention and deletion
AI and external services	Disclosed, not hidden	Settings page lists each service and what data it receives; AI-use flag stored on every version
Prompt injection	Uploaded text is data, never instructions	Input separation and output validation (see OWASP LLM risks)
Data use	No training on customer content	Provider terms and architecture both prevent it

7. Scope, feasibility and delivery plan

We have layered the plan on purpose: a small core that proves the idea from start to finish, and extras we can drop without breaking anything.

7.1 Scope

In scope for the MVP	Out of scope (stretch or later)
Upload and define guidelines, facts, terminology; owner, version, expiry Text, DOCX and PDF review Rules checks, claim matching, contradiction against past content Evidence cards, intent lock and suggested fixes Roles, comments, approvals, version history Claim Shelf-Life, Guideline Debugger (basic), Brand Passport (content hash), pre-mortem Redaction preview, audit log, AI-use disclosure, rules-only mode Basic image checks: alt text, contrast, palette	PPTX and advanced image review, logo matching Slack, Teams, Google Docs and CMS plug-ins Agent Gate API, Rule Simulator Voice Fingerprint, multilingual consistency Trust Report export Auto-publishing (a non-goal) Training models on customer content (a non-goal)

7.2 Delivery plan by stage and owner

Stage lengths will depend on how many days we have. The order and the exit checks stay the same.

Stage	Somnath (backend)	Falguni (frontend)	Om (ML)	Exit check
1. Foundations	Project setup, OpenAPI contract and API skeleton, auth and roles, MongoDB schema, file upload	UI shell, upload and source screens	Parsing, synthetic dataset and test set, retrieval index	Upload a source and see it indexed
2. Core loop	Review and findings endpoints, ML orchestration, findings storage	Draft submit screen, evidence card and comparison view	Claim extraction, matching and verdicts, rules checks	A draft returns sourced findings end to end
3. Workflow and unique features	Comments, approval and passport endpoints, versions, audit log, redaction gateway, Agent Gate API	Comments and approval UI, diff view, dashboard, intent lock	Fix suggestions with meaning check, pre-mortem, shelf-life logic	Full demo story runs for all roles
4. Hardening and demo	Security review, error handling, deployment	Polish, accessibility pass, demo flow	Evaluation report, threshold tuning, image checks	Metrics measured; demo rehearsed

7.3 Risks and fallbacks

Risk	Fallback
Model output is unreliable on tone or meaning	Keep tone advisory only; rely on exact rules and sourced checks; send low confidence to a human
Claim-to-source matching is noisy	Start with numbers, names and product terms; widen to free-text claims later
Image analysis is too weak in time	Limit visuals to alt text, contrast and palette
No real company data available	Use a fictional company, synthetic documents and public style guides
External model provider unavailable or not permitted	Rules-only mode and a small local model keep the core loop working
Integration between the Node backend and the Python service slips	Agree the API contract and internal ML endpoints in Stage 1 and mock them so the UI is not blocked

Scope creep	Stretch features are listed and ranked; none is needed for the demo

7.4 Why this is achievable with a three-person team

- Each person owns one layer with a clear contract, so we can work in parallel.
- Rules come first, so most findings are predictable and easy to test. The model only helps with matching meaning and drafting fixes.
- The MVP sticks to text, DOCX and PDF with basic image checks, and uses synthetic data we control.
- Everything is standard and well known: React, Node, MongoDB, FastAPI, FAISS and OpenCV.

7.5 Priority and effort by feature

Effort is a rough size: S is a day or less, M is two to three days, L is about a week of one person's time. We will rebalance if estimates prove wrong.

Feature	Priority	Effort	Owner
Upload sources, owners, versions, expiry	Must	M	Somnath
Rules checks (terms, superlatives, contrast, alt text)	Must	M	Om
Claim extraction, retrieval and matching	Must	L	Om
Evidence cards and comparison views	Must	L	Falguni
Roles, comments, approvals, versions	Must	L	Somnath, Falguni
Suggested fixes with Intent Lock	Must	M	Om, Falguni
Redaction preview, audit log, AI-use notice	Must	M	Somnath
Claim Shelf-Life	Should	S	Somnath
Brand Passport (content hash)	Should	S	Somnath
Pre-mortem	Should	M	Om
Insights dashboard and Guideline Debugger	Should	M	Somnath, Falguni
Basic image checks	Should	M	Om
Agent Gate API, Rule Simulator, Voice Fingerprint, multilingual check, Trust Report	Could	M to L	Shared

Cut line. If time runs short, we drop from the bottom of this table. The Must rows together are enough for the full demo story.

7.6 Definition of done for the MVP

- A user can upload sources, submit a draft with intent, and get findings that each cite a source passage.
- The returns-policy contradiction, the unsupported superlative and an expired fact are all caught in the demo.
- Creator, manager and reviewer roles can comment, approve and see version history.
- The UI shows a clear notice whenever AI or an outside service was used, and the tool works with the model switched off.
- We have measured recall, precision and citation rate on the test set and can show the numbers.

Appendix A. Alignment with the Inkloom AI problem statement

Problem statement requirement	How Preflight meets it	Owner
Import or define guidelines, approved facts, terminology, reusable assets	Knowledge base: upload PDF, DOCX or Markdown, or define items in the app; each has owner, version and expiry date	Somnath, Falguni
Review text, images, documents and other formats before publication	Text, DOCX and PDF in the MVP; PPTX and images (palette, logo, contrast, alt text) as the next step	Om, Somnath
Detect tone, terminology, messaging, facts, claims, visual and accessibility issues	Rules engine for exact checks; semantic claim-to-source matching; tone-drift check; WCAG contrast and alt-text checks	Om
Explain why it was flagged and name the guideline or source	Evidence card: the draft sentence, the proof passage (file, page, version) and a verdict with confidence	Om, Falguni
Suggest corrections that preserve the creator's purpose, without overriding humans	Intent lock: creator states purpose first; each fix shows a meaning check; fixes are opt-in	Om, Falguni
Comments, approvals, revision history, role-based review	Roles (creator, manager, reviewer, owner), threaded comments, approve or request changes, version diffs	Somnath, Falguni
Recurring problems and governance insights over time	Insights dashboard and Guideline Debugger: repeated mistakes, rules that are often overridden (likely unclear)	Somnath, Falguni
Protect confidential materials; disclose AI or external services	Tenant isolation, encryption, role-based access, redaction preview before any external call, visible AI-use notice, rules-only mode	Somnath, Om
Balance automation with human control; reliable, explainable results	No finding without a source; unproven items go to a human; every decision logged	All

Appendix B. Success metrics and demo

We plan to test on about 60 drafts for a made-up company, each with issues we planted ourselves. The targets are goals, and we will report what we actually measure.

Metric	Target	Measured by
Recall of planted issues	80% or higher	Found divided by planted
Precision of findings	85% or higher	Reviewer agreement
Findings with a cited source	100%	Automated check on every report
Self-contradictions caught	80% of planted ones	Planted vs found
Review time per draft	30% lower than manual	Timed pilot, 5 users
Unclear rules identified	At least 3	High override or dispute rate

Demo story: a creator writes "Free returns within 14 days". Preflight puts that next to the uploaded policy (page 3, version 2, which says 30 days), marks it Contradicted and suggests a fix that keeps the message. The manager approves it and a Brand Passport is saved. Another post says "fastest in India", but we have no evidence for it, so it goes to legal as a question instead of being called wrong. After 50 drafts the dashboard shows one rule being overridden 70% of the time, which tells the owner it needs clearer wording.

Appendix C. Team roles

Member	Role	Owns
Somnath	Team Lead, MERN Full Stack Developer	Backend and APIs: Node.js, Express, MongoDB, REST API and OpenAPI contract, integrations and Agent Gate API, auth and roles, workflow, audit log, security, redaction gateway, ML orchestration, insights aggregation
Falguni Sharma	MERN Full Stack Developer	Frontend (React): review UI, evidence cards, comparison views, comments and approval UI, dashboard, accessible and responsive interface
Om Apar	AI/ML Specialist	Machine learning: claim extraction, retrieval, matching, tone and contradiction models, intent check, pre-mortem, OCR and image checks, evaluation